

Role of Adsorption-Induced Deformation on Gas Self-Diffusivity in a Flexible Microporous Coal Matrix

Quanlin Yang*, Junhua Xue, Haifei Lin, and Zhehui Jin*



Cite This: *Langmuir* 2025, 41, 10971–10981



Read Online

ACCESS |



Metrics & More

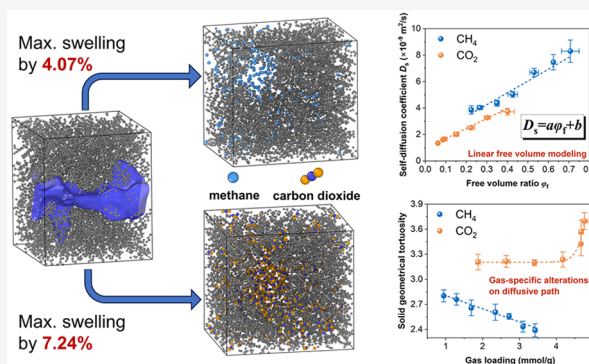


Article Recommendations



Supporting Information

ABSTRACT: Adsorption-induced deformation has long been underappreciated in gas transport studies of microporous coal, yet it strongly influences pore configurations and diffusive pathways. Here, a hybrid grand canonical Monte Carlo (GCMC)/molecular dynamics (MD) approach and equilibrium MD (EMD) simulations are employed to investigate how matrix flexibility reshapes pore structures and, in turn, impacts CH₄ and CO₂ self-diffusion in connected pore networks under various gas loadings. The results show that coal matrix deformation enhances adsorption, with CO₂ exhibiting greater uptake and volumetric strain than CH₄. A universal linear relationship emerges among gas loading, free volume ratio, and self-diffusion coefficients for both rigid and flexible matrices. In flexible matrices, this linearity features a gentler slope, indicating reduced diffusion sensitivity to diminishing free volume with loadings. By comparing geometrical and effective tortuosity, it is revealed that strongly adsorbing CO₂ induces significant swelling and complex local rearrangements at elevated loadings, pushing geometrical tortuosity far beyond rigid-matrix levels, whereas CH₄—with weaker adsorption—drives smaller, more uniform structural adjustments that only mildly increase geometrical tortuosity. These differences in tortuosity directly reflect changes in path complexity, which in turn governs self-diffusion behavior. Collectively, the findings clarify the dynamic coupling between gas adsorption, matrix deformation, and self-diffusivity in microporous coal, offering critical guidance for enhanced gas recovery and CO₂ sequestration strategies that rely on accurate modeling of gas transport in deformable media.



1. INTRODUCTION

Gas transport in nanoporous coal media is vital for optimizing coalbed methane (CBM) extraction and geological carbon sequestration (GCS), directly impacting both CH₄ recovery efficiency and CO₂ injectivity.^{1,2} Conventional continuum models, such as the Darcy's law, which assumes viscous flow, are inadequate at the nanoscale due to significant gas adsorption and the breakdown of classical hydrodynamics frameworks.^{3,4} Similarly, experimental methods based on the Fick's law provide limited insights,^{5,6} as they rely on assumptions (e.g., isotropic and homogeneous solid structure, absence of deformation, continuity principle) that fail to capture the complexity and deformability of coal. Numerous experimental and simulation studies have demonstrated that gas adsorption induces coal deformation, a critical phenomenon that might significantly influence gas transport in coal.^{7–13} Gas is predominantly stored within micropores of the coal matrix,^{14,15} where molecular transport is governed by diffusivity rather than bulk flow.^{16–18} In ultraconfined spaces typical of coal micropores, collective effects can be neglected in the long-term diffusion, resulting in the convergence of self- and collective diffusion coefficients.^{19–22} Consequently, focusing on self-diffusivity is justified for understanding gas storage and transport mechanisms in this context. Further-

more, boundary-driven nonequilibrium molecular dynamics (BD-NEMD) simulations have shown that CH₄ mass transfer coefficients align closely with the Onsager coefficients under equilibrium conditions, confirming that transport within the microporous matrix is inherently diffusive and that transport coefficients can be reliably derived from equilibrium molecular dynamics (EMD) simulations.²³ Therefore, a molecular-level understanding of gas self-diffusivity in a deformable microporous coal matrix is essential for improving the accuracy of predictions in gas extraction and storage.

Within the EMD framework, numerous studies have explored the impact of the pore structure characteristics on gas diffusion. For example, Thyagarajan *et al.*²⁴ reported that self-diffusion coefficients of CH₄ and CO₂ in rigid amorphous nanoporous materials initially increase with adsorbate concentration, peak, and then decrease due to steric hindrance,

Received: January 24, 2025

Revised: April 1, 2025

Accepted: April 10, 2025

Published: April 22, 2025



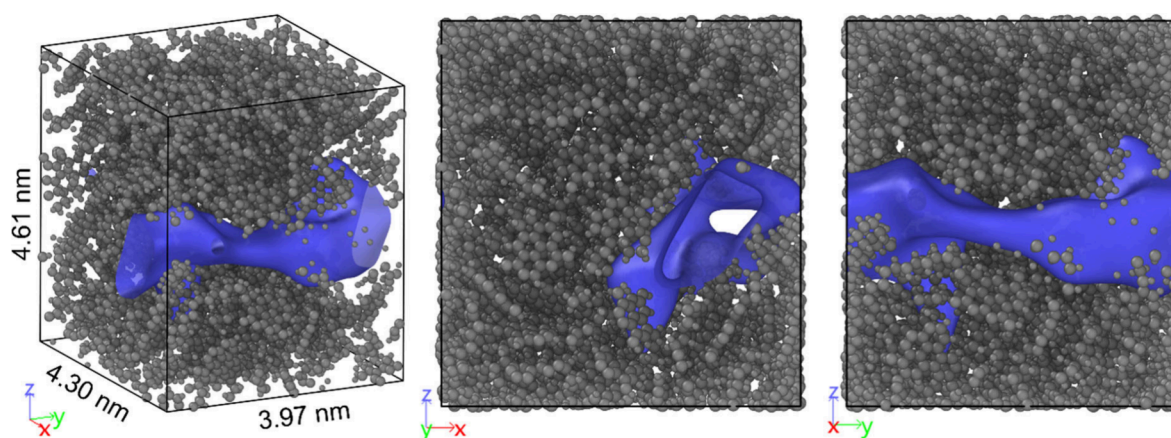


Figure 1. Schematic representation of the molecular model of the coal matrix framework (gray spheres) and connected pore network (transparent blue surface), which forms a continuous diffusive path. The surface mesh is constructed using a Gaussian density method in the open visualization tool (OVITO Basic, version 3.10.6).³⁹ For clarity, all solid atoms are depicted in gray, and closed pores are omitted to emphasize the connected pore network.

showing no clear correlation with structural descriptors such as limiting pore diameter (LPD). Yu *et al.*²⁵ reported anisotropy in CH₄ self-diffusivity, which positively correlates with pore volume in type II kerogens. Aljaberi *et al.*²⁶ observed a positive correlation between CH₄ self-diffusion and porosity but a negative correlation with kerogen tortuosity. Conversely, Vasileiadis *et al.*²⁷ identified a linear relationship between the largest component of the self-diffusion coefficient and LPD in type II-D kerogen structures, while accessible pore volume and surface area show weaker correlations. In our recent study,²⁸ by distinguishing connected pores from unconnected ones, we revealed the relationship between CH₄ self-diffusion and key pore informatics, including pore connectivity, porosity, LPD, and diffusive tortuosity. However, these studies largely overlook the poromechanical coupling between adsorbed fluids and flexible solid structures. Given coal's deformable nature, gas adsorption induces structural changes that alter pore morphology and topology—such as pore connectivity, porosity, pore size distribution (PSD), and tortuosity¹²—and ultimately modify the diffusion landscape.

On the other hand, Wu *et al.*²⁹ used BD-NEMD simulation to show that CH₄ transport diffusivity is generally lower in flexible kerogen than in rigid kerogen, in spite of occasional increases from pore opening. However, their results might be dependent on mechanical constraints inherent to the BD-NEMD method, such as fixed system size and frozen atoms, which restrict kerogen swelling.¹⁹ Since transport properties in NEMD can be sensitive to applied forces, EMD is often preferred. Using a sequential “first-GCMC-then-MD” approach,³⁰ Obliger and his co-workers^{18,21,31} applied the Fujita–Kishimoto free volume theory to relate fluid loading, free volume, and diffusion, concluding that CH₄/CO₂ adsorption-induced kerogen swelling increases free volume and enhances gas diffusion. However, this approach cannot capture the dynamic coupling between adsorption and matrix deformation,¹² potentially leading to inaccurate descriptions on the effect of coal deformation on gas diffusion. In contrast, a hybrid grand canonical Monte Carlo/molecular dynamics (GCMC/MD) simulation performed by the same research group under drained/unjacketed conditions showed that argon (Ar) self-diffusivity in microporous carbon decreases with increased fluid loading due to the reduced free volume.³² This positive correlation between self-diffusivity and free volume is

also observed in amorphous poly(alkyl acrylates).³³ Although these trends in self-diffusion with fluid loading are sometimes contradictory, likely due to different simulation methods, the aforementioned studies consistently indicate that matrix swelling and internal structural deformation improve pore connectivity, enhancing fluid diffusion in flexible matrices compared with rigid structures.

In many recent studies on fluid diffusion in microporous media, the GCMC algorithm is commonly used to insert fluid molecules into the matrices, thereby generating the initial configurations for diffusion studies. This approach unavoidably results in some molecules being inserted into closed or unconnected pores. However, subsequent calculation of self-diffusion coefficient typically includes all molecules, regardless of their actual mobility or displacement, leading to underestimated diffusivities, as trapped molecules contribute little to diffusivity, especially in the long-time limit characteristic of the Fickian diffusion regime.^{4,28} This study resorts to hybrid GCMC/MD and EMD simulations to focus on CH₄ and CO₂ diffusion in connected pore networks of rigid and flexible matrices. We aim to investigate the influence of matrix flexibility on gas diffusion by analyzing pore structure evolution, applying free volume theory, assessing tortuosity metrics, and examining gas-specific matrix deformation effects on diffusion pathways. Utilizing molecular simulation techniques, this study offers deeper insights into gas transport behaviors in coal, ultimately improving the ability to model the CBM recovery and GCS processes.

2. MATERIALS AND METHOD

2.1. Molecular Models. The bulk coal matrix configurations are modeled using type III-A kerogen (C₂₃₃H₂₀₄O₂₇N₄, molecular weight $M_w = 3492.12$ g/mol) based on the work of Ungerer *et al.*,³⁴ representative of bituminous coals,³⁵ chosen for its alignment with the chemical composition, maturity, and microporous structure of the coal, making it more suitable for the geological conditions of this study. To ensure meaningful self-diffusion calculations, the coal matrix models must feature LPDs (defined as the maximum size of a probe capable of traversing the simulation cell along a continuous pathway in at least one dimension) that exceed the kinetic diameters of CH₄ (0.38 nm) and CO₂ (0.33 nm), as models with an unconnected pore network would otherwise yield zero self-diffusivity in the long-time limit.^{4,24} Therefore, initial coal matrix models are constructed by randomly placing 15 III-A monomers and several dummy particles

(DPs) within a 10 nm × 10 nm × 10 nm cubic simulation box using the Packmol package,³⁶ as illustrated in Figure S1 in the Supporting Information (SI). The DPs are systematically optimized through extensive testing of various combinations of Lennard-Jones (LJ) parameters and numbers until a connected pore network with an appropriate LPD is successfully formed. The final parameters chosen are an LJ diameter (σ_{DP}) of 1 nm, an LJ energy parameter (ϵ_{DP}) of 0.0199 kcal/mol (comparable to that of a hydrogen atom), a mass (M_{w}) of 40 g/mol, and total DP number of 12. These values ensure that the DPs serve as effective geometric placeholders to establish a stable pore network.^{27,37,38} The system then undergoes a staged relaxation process involving a series of canonical (NVT) and isothermal–isobaric (NPT) MD simulations to thoroughly explore the coal matrix's free energy landscape.¹² Subsequently, all DPs are removed, and a 20 ns NPT MD simulation is performed at 313.15 K and 0 bar to achieve final equilibration of the interconnected coal matrix, as summarized in Table S1. The resulting connected pore network is depicted in Figure 1.

The consistent valence force field (CVFF),^{40,41} TraPPE-UA,⁴² and TraPPE-small⁴³ models are employed to simulate the III-A macromolecule, CH₄ and CO₂ molecules, respectively. CO₂ is maintained rigid using a rigid-body algorithm.⁴⁴ A cutoff distance of 1.4 nm is applied for both 12–6 LJ and coulombic pairwise interactions. In EMD simulations (NVT/NVE ensembles), where the atom count and cell volume remain constant, analytical corrections are applied to estimate the contributions from interactions beyond the cutoff.⁴⁵ In contrast, for hybrid GCMC/NPT simulations, characterized by dynamic changes in atom numbers and box dimensions, a shifted potential approach is used to ensure that the LJ potential smoothly decays to zero at the cutoff. This strategy guarantees a consistent treatment of long-range interactions and accurate thermodynamic properties across different simulation conditions. The particle–particle particle-mesh (PPPM) method, with a precision of 1×10^{-5} , is used for long-range coulombic interactions. Pairwise energy and force contributions are excluded for atoms directly bonded or separated by one or two intermediate bonds. The LJ diameter (σ_{LJ}) and energy (ϵ_{LJ}) parameters for interactions between dissimilar atoms are determined using Lorentz–Berthelot mixing rules. Temperature and pressure are regulated using a Nosé–Hoover thermostat and an anisotropic barostat, with characteristic damping factors of 0.1 and 1 ps, respectively. All molecular simulations are conducted under three-dimensional periodic boundary conditions (PBCs) using the LAMMPS software package (version 29 Aug 2024) with a time step of 1 fs.⁴⁶

The constructed coal matrix exhibits a density of 1.10 g/cm³. This value is slightly lower than those reported for bituminous coal in previous molecular simulations (1.17–1.24 g/cm³)^{12,34} and experimental studies (1.22–1.25 g/cm³).^{47,48} The modest deviation is acceptable because DPs are introduced to generate a connected pore network, thereby increasing the overall matrix volume. Self-diffusion coefficients of bulk CH₄ and CO₂ obtained from molecular simulations using the TraPPE force field are compared with available NMR experimental measurements, as shown in Figure S2. Good agreement between our simulation and experiment is observed for both gases.

2.2. Hybrid GCMC/MD Simulation. To create fluid-saturated structures that account for coal matrix flexibility, a hybrid GCMC/MD approach is used in LAMMPS, fully capturing the coupled effects of gas adsorption and matrix deformation.⁴⁹ This method alternates gas insertion/deletion MC attempts with NPT MD steps, maintaining thermal, chemical, and mechanical equilibrium between the system and an infinite gas reservoir. This setup mimics experimental conditions where coal is in a bulk gas reservoir under zero effective stress (*i.e.*, solid confining pressure equals bulk gas pressure).³² The chemical potentials for the GCMC simulations, at a temperature of 313.15 K and pressures ranging from 5 to 300 bar, are determined using the Widom insertion method⁵⁰ in the NVT ensemble via the MCCCSTowhee (v8.2.x) package.⁵¹ These values are further validated through GCMC simulations in an initially empty box. Each hybrid GCMC/MD simulation comprises 30,000 blocks for

CH₄ and 40,000 blocks for CO₂ to ensure convergence, with each block consisting of 1000 GCMC attempts (with a 50% insertion/deletion probability) followed by 1000 NPT MD steps. The first half of the blocks is for equilibration, while the remaining half is used for sampling. Additional simulations are performed by using a rigid coal matrix for comparison.

2.3. Equilibrium MD Simulation. Self-diffusivity of CH₄ and CO₂ along the connected pore network is investigated as shown in Figure 1 using EMD simulations. For each equilibrated hybrid GCMC/MD trajectory during the production period, the configuration with the number of gas molecules closest to the ensemble average is selected as the initial configuration for EMD simulations. The self-diffusion coefficient is then calculated following a rigorous protocol, as detailed in refs 52 and 53:

- (1) A 10 ns NPT MD simulation is performed on the selected initial configuration to sample the system's average volume.
- (2) The final configuration from the NPT MD simulation is scaled to this average volume and used as the starting point for a 10 ns NVT MD thermalization run.
- (3) During the subsequent 50 ns NVT MD production run, the coal matrix is kept frozen while the gas molecules are simulated at a temperature of 313.15 K. The approach of immobilizing solid atoms minimizes systematic errors arising from the drift of the system's center of mass (COM) and prevents gas in closed pores from being artificially displaced by matrix movements, thereby ensuring that the measured self-diffusion coefficients accurately reflect the intrinsic mobility of gas molecules.

An integration time step of 2 fs is used, and configurations for mean squared displacement (MSD) calculations are saved every 1 ps. While NVE MD simulations are typically recommended for sampling particle trajectories,^{4,52–54} CO₂ diffusion presents a unique challenge. The slow convergence of electrostatic interactions (with a 1.4 nm cutoff distance) results in Hamiltonian nonconservation and a gradual temperature increase. As demonstrated by Hammonds and Heyes,⁵⁵ even minor inaccuracies in electrostatic convergence can accumulate over long simulation time, leading to measurable temperature drift. To mitigate this effect and ensure accurate diffusion measurements, we weakly coupled CO₂ molecules to a Nosé–Hoover thermostat in the NVT ensemble with a damping factor of 1 ps.

The self-diffusion coefficient D_s is calculated using the Einstein relation,

$$D_s = \frac{1}{2dN} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle \sum_{i=1}^N [r_i(t) - r_i(0)]^2 \right\rangle \quad (1)$$

where d is the spatial dimension of the system, taken as 1 in this study, as the diffusive pathway is confined along the y -direction (Figure 1); N represents the number of gas molecules; $r_i(t)$ denotes the position of molecule i at time t . Equation 1 is valid for time scales where the slope of the MSD versus time plot reaches 1 on a log–log scale,^{54,56} as shown in Figure S4. This criterion serves as a guideline for determining the minimum simulation duration required to accurately compute D_s using the Einstein relation.

In our recent study,²⁸ it is emphasized that distinguishing between connected and unconnected pores is crucial for understanding gas transport in rigid kerogen structures. Accordingly, gas molecules located in connected pores are labeled as “free gases”, while those in unconnected pores are deemed “trapped gases”. Connected pores form continuous pathways (at least in one direction, aided by PBCs), allowing free gases to exhibit a steadily increasing MSD over time. To identify this behavior, we perform 10 ns NVE MD simulations, saving atomic positions at regular intervals. The MSD for each gas molecule is then computed in LAMMPS. Molecules whose MSD values continue to grow far beyond the square of the largest pore diameter (~ 1.5 nm) are classified as “free”. Conversely, molecules whose MSD values saturates below this threshold are deemed “trapped”. This approach ensures that only free gases are considered when the

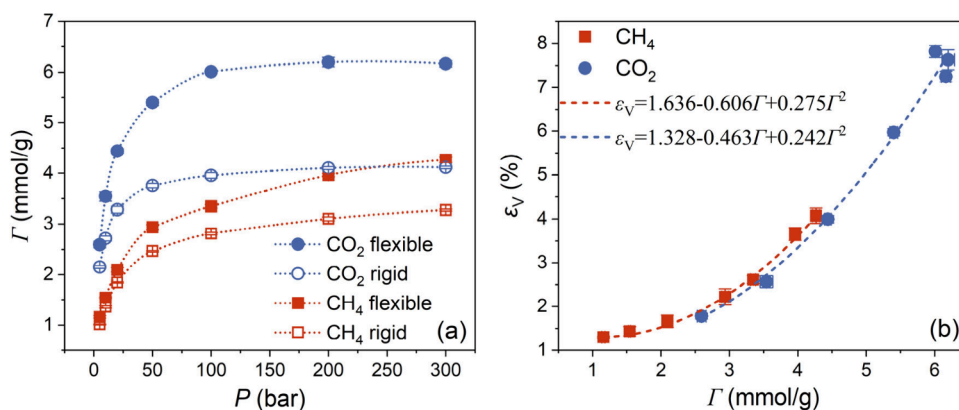


Figure 2. (a) Adsorption isotherms of CH₄ and CO₂ in rigid and flexible coal matrices. Dotted lines are included for visual guidance. (b) Volumetric strain of flexible coal matrices as a function of gas loading. Most error bars are smaller than the symbols.

Einstein relation is applied to compute D_s in flexible and rigid coal matrices.

Consistent with previous studies on gas self-diffusivity in porous solids,^{4,23,25,27,31,32} the simulation protocols employed herein did not incorporate explicit finite-size corrections.^{22,53} Validation against experimental data (Figure S2) confirms that any systematic offsets attributable to finite-size effects are minimal and do not compromise the relative trends observed. This limitation is acknowledged, and future investigations will address larger system sizes and analytic corrections to further refine these estimates.

2.4. Accessible Diffusional Pore Space and Tortuosity. The pore space available for the dynamics of CH₄ and CO₂ molecules within the coal matrix is analyzed using PoreBlazer v4.0, a software package developed by Sarkisov *et al.*⁵⁷ This study focuses on the network-accessible space, defined as the region a specific probe molecule can reach and traverse through a continuous, physically meaningful pathway that accounts for the probe's size. The pore volume is determined as the space occupiable by the probe, enclosed by the Connolly surface. Spherical probes with diameters of 0.373 nm for CH₄ and 0.330 nm for CO₂ are used in this analysis. It should be noted that the LPD, which represents the maximum diameter of a spherical probe that can navigate the pore network, is predominantly determined by the intrinsic geometry of the coal matrix and, therefore, remains relatively invariant between CH₄ and CO₂ probes under the initial conditions prior to adsorption. In contrast, parameters such as accessible pore volume (APV), accessible porosity (AP), and specific surface area (SSA) are more sensitive to gas-specific interactions and the effective probe sizes used, thereby exhibiting noticeable variations between the two gases, as shown in Figure 5.

Tortuosity is a fundamental parameter reflecting pore network complexity, significantly influencing diffusion resistance in porous materials.^{37,58,59} Traditionally, tortuosity is discussed broadly within the context of transport phenomena, often without clearly distinguishing structural effects from molecular interaction effects. This distinction is particularly crucial in microporous coal matrices due to the strong influence of gas–solid interactions on the diffusion behavior. In this study, two types of tortuosities are distinguished: geometrical tortuosity (τ_g) and effective tortuosity (τ_e). τ_g quantifies structural influences alone, and it isolates the intrinsic geometrical constraint on diffusion pathways, defined as

$$\tau_g = \sqrt{\frac{D_0}{D_{sg}}} \quad (2)$$

where D_0 is the self-diffusion coefficient of the gas in the bulk phase at the same temperature and density as in the confined pore space; D_{sg} is the self-diffusion coefficient of the gas within the pore network when all attractive interactions are removed. Notably, D_0 is not constant, which decreases with increasing gas density as more frequent gas–gas collisions slow the overall diffusion. Consequently, the ratio D_0/D_{sg} compensates for gas–gas interactions, ensuring that τ_g reflects only

the structural influence of the solid on diffusion behavior. Specifically, gas–solid attractions are eliminated by applying a reduced cutoff distance of $2^{1/6}\sigma$ to the LJ potential, retaining only repulsive forces. Additionally, coulombic interactions between CO₂ and III-A macromolecules are disabled by setting solid atom charges to zero, thus eliminating electrostatic interactions between CO₂ and the solid matrix while preserving intermolecular interactions among CO₂ molecules. This modification ensures that τ_g exclusively captures the geometric influence of confined pore structures on gas diffusion. Consequently, variations in τ_g reveal how evolving pore geometry impacts diffusion pathways over time.

On the other hand, τ_e accounts for both the structural geometry of the porous medium and the gas–solid interactions, defined as²⁰

$$\tau_e = \sqrt{\frac{D_0}{D_s}} \quad (3)$$

τ_e quantifies the extent to which a porous medium impedes gas diffusion compared with free space, arising from a combination of structural complexity and gas–solid interactions. Unlike τ_g , which is purely structural, the additional gas–solid interactions can transiently trap gas molecules near the pore walls, resulting in intermittent Brownian motion that influences the overall self-diffusivity. This stop-and-go diffusion pattern, characterized by alternating surface adsorption and in-pore diffusion, has been extensively documented in studies of gas and water diffusion in amorphous materials.^{4,60,61} To illustrate this, we simulate the self-diffusion of four single CH₄ molecules in a coal matrix and compare the results with Bousige *et al.*'s work on simple LJ fluid diffusion in disordered media,⁴ as shown in Figure S3. Although individual trajectories exhibit intermittent motion, ensemble averaging leads to a Fickian diffusion regime with an effective self-diffusivity, as shown in Figure S4. Thus, while gas molecules in microporous coal matrices inherently display intermittent Brownian motion, our study emphasizes how coal matrix deformation affects τ_e and consequently influences the overall gas self-diffusivity.

To address the inherent uncertainties of the amorphous coal matrix, statistical reliability is ensured by calculating error bars for matrix structural and gas diffusional properties using the standard error of the mean across five to six independent configurations and simulation runs. For other quantities, such as gas loading and volumetric strain, error bars are determined through ensemble averaging during the production run.

3. RESULTS AND DISCUSSION

3.1. Gas Adsorption in Deformable Coal Matrices.

Figure 2 demonstrates the coupled process of gas adsorption and coal matrix deformation and its potential impact on gas diffusion. The adsorption amounts of CH₄ and CO₂ are consistently higher in flexible matrices than those in rigid ones

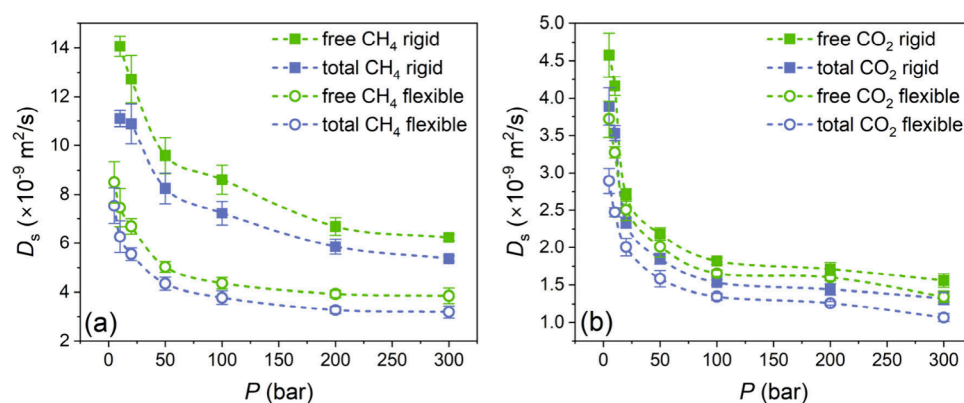


Figure 3. Influence of pore connectivity and matrix flexibility on the D_s of (a) CH_4 and (b) CO_2 . The terms “rigid” and “flexible” in the legends indicate the matrix flexibility, while “free” and “total” denote D_s in connected pores and all pores, respectively. Dashed lines are included for visual guidance.

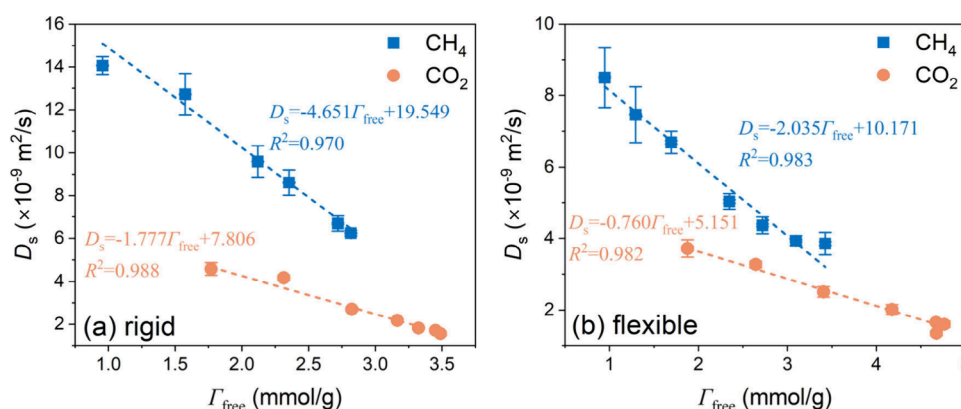


Figure 4. D_s of CH_4 and CO_2 as functions of free gas loading (Γ_{free}) in (a) rigid and (b) flexible coal matrices. Short dashed lines represent linear fitting results.

under all pressure conditions, highlighting that deformation enhances pore availability. CO_2 exhibits a more significant increase in the adsorption amount due to its stronger interactions with the matrix, driven by the quadrupole moment and its ability to access narrower micropores. Figure 2(b) depicts a quadratic relationship between volumetric strain and gas loading, with CO_2 causing greater swelling than CH_4 under the same pressure. This coupling between adsorption and deformation suggests that matrix flexibility drastically alters the pore structure, directly influencing diffusion pathways. These findings underscore the importance of considering gas-specific interactions and matrix flexibility in diffusion studies.

3.2. Gas Diffusion in Connected Pore Networks. To emphasize the role of pore connectivity in fluid transport within amorphous structures,²⁸ D_s of all gas molecules and those residing in connected pore networks are compared, as shown in Figure 3. Including molecules trapped in unconnected pores leads to an underestimation of the D_s . For CH_4 , this reduction ranges from 12.4% to 21.1% in rigid matrices and 13.3% to 18.8% in flexible matrices, respectively; for CO_2 , the reduction spans 13.8% to 15.9% in rigid matrices and 18.6% to 24.4% in flexible matrices, respectively. These reductions correspond to the number fraction of “trapped gases” relative to the total gas loading. Consequently, all subsequent analyses are based on the D_s of these “free gases”.

As the pressure increases, D_s decreases consistently across all matrix types and gas species. This decline reflects the constraints imposed by denser fluid conditions, which limit

molecular mobility either by reducing the overall space available for movement or by increasing intermolecular collisions that hinder diffusion process. The consistent downward trend with increasing pressure underscores the sensitivity of diffusion to the overall molecular packing conditions and the distribution of accessible free volume within the matrix, which will be discussed in the following sections.

Although both CH_4 and CO_2 exhibit decreasing D_s with increasing pressure, matrix flexibility, and gas loading, as shown in Figure 4, CH_4 consistently achieves higher diffusivities across all examined conditions compared to CO_2 . This disparity arises from their distinct molecular characteristics: CH_4 , being smaller and nonpolar, primarily diffuses under constraints imposed by pore dimensions and connectivity, whereas CO_2 's linear geometry and partial charges promote stronger gas–matrix interactions, which significantly influence its transport. These observations underscore that even under similar pore environments and gas densities, intrinsic molecular properties can yield remarkably different diffusion behaviors.

Contrary to previous studies suggesting that structural flexibility promotes molecular motion,^{18,21,31} results in this work indicate that gas molecules diffuse more slowly in flexible matrices than in rigid ones. As shown in Figure 5, rising pressure (and thus gas loading) causes the flexible matrix to develop a higher APV, AP, and SSA compared to its rigid counterpart. Although these increases do not raise D_s above

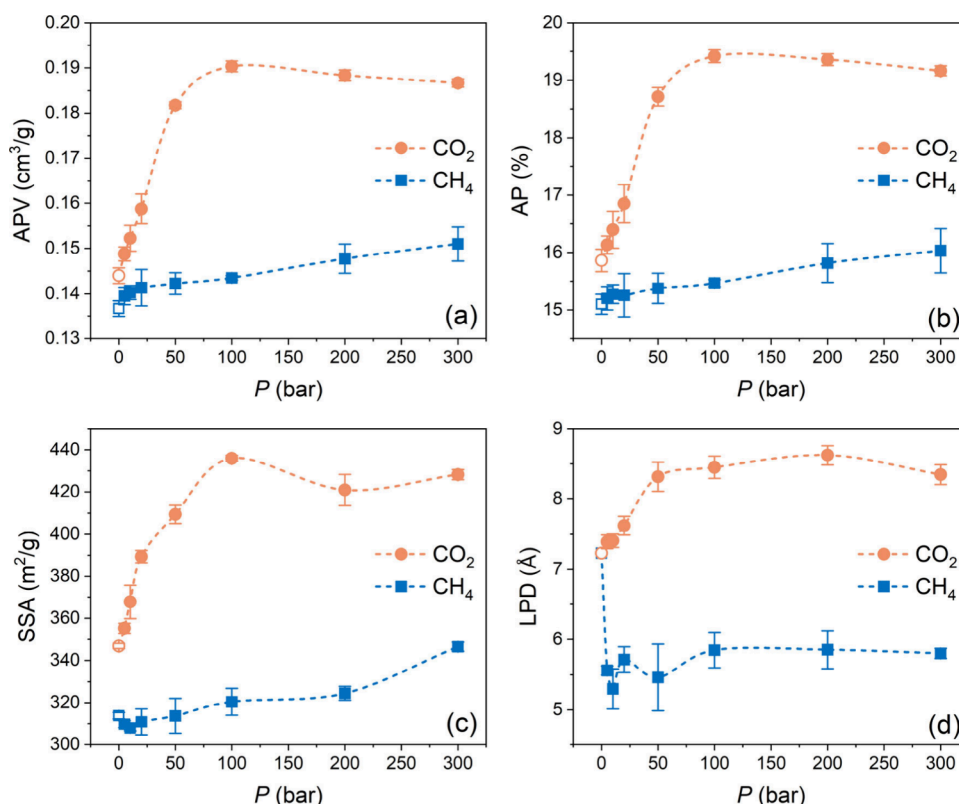


Figure 5. Evolution of the pore structure parameters with increasing pressure for CH_4 and CO_2 in flexible matrices. Hollow symbols represent the initial parameters before gas adsorption (rigid matrix), and differences between CH_4 and CO_2 are attributed to distinct pore characterization probe sizes, as detailed in Section 2.4. Short dashed lines are provided for visual guidance.

those in the rigid matrices, they mitigate the decline in diffusion with increasing loading, as evidenced by the flatter slope of D_s versus Γ_{free} in flexible matrices shown in Figure 4(b). In other words, although flexible matrices never surpass rigid matrices in terms of absolute diffusivity at any given pressure, their structural adaptability under dense conditions alleviates the severity of diffusion reduction. This adaptability, driven by the flexible matrix's ability to swell and adjust its internal structure, may lessen the negative impact of higher loadings on gas diffusion.

Figure 5(d) highlights the contrasting trends in the LPD for CH_4 and CO_2 , with little apparent correlation to their D_s . For CH_4 , weaker gas–solid interactions limit matrix deformation, as the confining pressure in GCMC/NPT simulations curbs significant swelling that might otherwise reduce LPD. By contrast, stronger CO_2 –matrix interactions lead to more obvious adsorption-driven swelling (Figure 2(b)), preferentially expanding pathways to accommodate its larger molecular size (Figure 5(a)). This swelling effectively counters confining pressure, preserving larger continuous pathways essential for CO_2 diffusion and thereby increasing LPD. Although the relationship between LPD and D_s remains inconclusive in the literature,^{24,27,28} these findings suggest that LPD and D_s are not strictly correlated or at least that LPD alone is not the sole determinant of self-diffusion.²⁸

3.3. Universal Linear Free Volume Modeling. As illustrated in Figure 4, a strong correlation exists between D_s and Γ_{free} , indicating the necessity of explicitly considering density effects. Effective diffusion can occur only if gas molecules encounter sufficiently large voids within their immediate surroundings. Therefore, the free volume

theory^{3,16–18,21,28,30–32,62} has been widely adopted in previous studies to quantitatively describe gas diffusion behavior in amorphous porous solids. This theory relies upon the probability that a gas molecule finds a neighboring cavity larger than a certain critical volume ν_{crit} . Thus, D_s can be expressed as

$$D_s = D_s^0 \exp\left(-\frac{N\nu_{\text{crit}}}{V_{\text{free}}}\right) \quad (4)$$

where ν_{crit} represents the minimal cavity volume necessary for gas molecule diffusion, N is the number of gas molecules, and V_{free} denotes the available free volume for gas molecular motion. V_{free} is defined as the difference between the total accessible pore volume and the volume occupied by the gas molecules. Approximating ν_{crit} by the volume of a single gas molecule, ν' , and expressing V_{free} as

$$V_{\text{free}} = V_{\text{free}}^0 - N\nu' \quad (5)$$

where the initial free volume V_{free}^0 is interpreted as the total accessible volume of the connected pore network without any gas molecules adsorbed. Consequently, eq 4 transforms to

$$\log\left(\frac{D_s}{D_s^0}\right) \propto -\frac{N\nu'}{V_{\text{free}}} = 1 - \frac{V_{\text{free}}^0}{V_{\text{free}}} \quad (6)$$

In this context, D_s^0 denotes the self-diffusion coefficient at an infinitely small gas loading. Introducing the dimensionless free volume ratio $\phi_f = V_{\text{free}}/V_{\text{free}}^0$, eq 4 further simplifies to

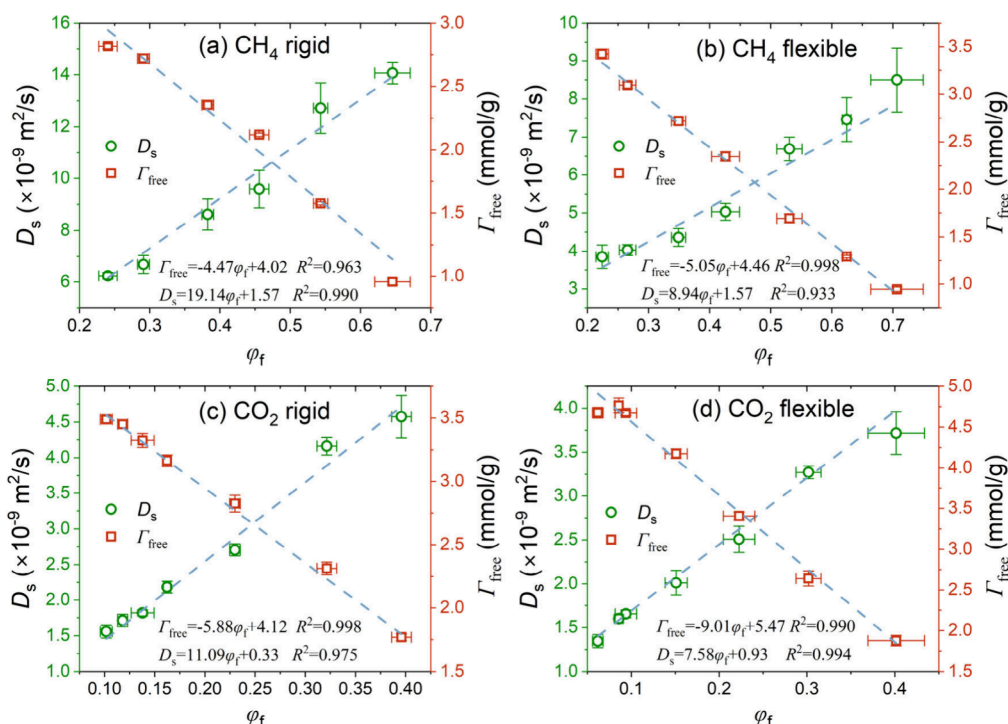


Figure 6. Linear relationships between the free volume ratio (ϕ_f), self-diffusion coefficient (D_s), and free gas loading (Γ_{free}) for CH_4 and CO_2 in rigid and flexible matrices. Dashed lines represent the linear fitting results.

$$D_s \propto D_s^0 \exp\left(1 - \frac{1}{\phi_f}\right) \quad (7)$$

where ϕ_f can be calculated using the PoreBlazer software by comparing accessible probe-occupiable pore volumes with and without gas molecules adsorbed.

The free volume theory identifies the availability of sufficient pore space as the critical determinant for gas molecule mobility. To validate the applicability of the free volume theory for coal– CH_4 systems, eq 7 was employed to establish a correlation between D_s and $1 - 1/\phi_f$. The obtained results are illustrated in Figure S5, confirming that the theory effectively captures diffusion behaviors for various gases in different matrix frameworks, with correlation coefficients (R^2) consistently exceeding 0.98. According to these fitted equations, as ϕ_f approaches 1 (*i.e.*, the scenario of infinitesimally low gas density), D_s approaches D_s^0 . The extrapolated D_s^0 values closely match those from Figure 4, as the gas loading approaches zero, and the detailed comparison is summarized in Table S2.

Furthermore, a simpler linear relationship can also effectively characterize the interplay among ϕ_f , Γ_{free} , and D_s , as shown in Figure 6. Unlike prior studies that report an exponential dependence of D_s on ϕ_f ,^{3,16,17,28} results in this work consistently demonstrate a clear linear relationship. This suggests that gas diffusion in these systems is governed by smooth and proportional changes in the free volume ratio. The observed linearity is strikingly universal, applicable to both CH_4 and CO_2 in rigid and flexible matrices. Thus, gas diffusion can be effectively captured by a simple linear free volume model, providing a practical and accurate alternative to widely used exponential formulations. For instance, in our previous study, where the D_s – ϕ_f relationship is fit to an exponential equation,²⁸ the general trend also closely approximates a linear relationship.

Figure 6 highlights the significant influence of matrix deformation on gas diffusion. In flexible matrices, structural adjustments occur with increasing Γ_{free} , resulting in higher accessible pore volumes, as previously shown in Figure 5. These structural adaptations mitigate the decline in diffusion that is typically caused by gas densification under high loadings. In contrast, rigid matrices exhibit a linear relationship between D_s and ϕ_f directly reflecting static and intact void spaces, as no structural compensation takes place. Flexible matrices, however, display flatter slopes in the D_s – ϕ_f relation (see Figure 6(b) and (d)), indicating that deformation maintains sufficient free volume, effectively buffering the diffusion decline. While deformation does not increase diffusion beyond levels observed in rigid matrices, it slows the rate of decline, preserving molecular mobility under high-loading conditions.

Moreover, D_s exhibits distinct gas-specific behaviors. CH_4 consistently displays a steeper slope in the D_s – ϕ_f relation compared to CO_2 in both rigid and flexible matrices, indicating a higher sensitivity to free volume changes. This behavior arises from the smaller size of CH_4 and its weaker interaction with the solid, enabling it to navigate constricted pathways more efficiently. In contrast, CO_2 shows a weaker diffusion enhancement with increasing ϕ_f largely due to its stronger interaction with the coal matrix and dependence on larger pore spaces for mobility. This is further evidenced by the higher Γ_{free} values for CO_2 at comparable ϕ_f reflecting its stronger affinity to the coal matrix. Matrix flexibility mitigates the decline in D_s for both gases by maintaining an accessible free volume under increased loading. However, the impact of flexibility is more pronounced for CH_4 , as the D_s – ϕ_f slope drops by >50% from rigid to flexible matrices, benefiting from its weaker interactions with the coal matrix. For CO_2 , the benefits of flexibility are partially counteracted by its stronger adsorption, leading to an approximately 30% reduction in slope.

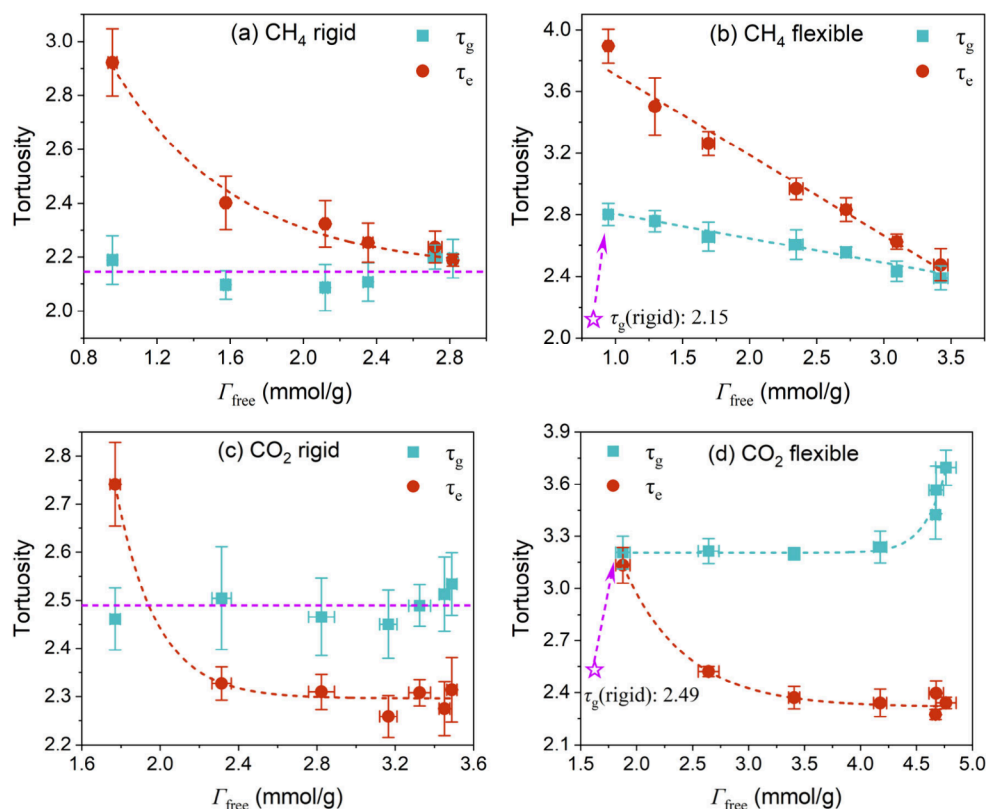


Figure 7. Geometrical (τ_g) and effective (τ_e) tortuosities of rigid and flexible coal matrices across varying gas loadings. Pink dashed lines in (a, c) indicate the average τ_g in rigid matrices, while cyan and red dashed lines provide visual guidance for tortuosity trends. Hollow stars in (b) and (d) mark the initial τ_g in flexible matrices.

3.4. Impact of Tortuosity in Gas Diffusion within Deformable Matrices. Figure 7 shows the variations of geometrical (τ_g) and effective (τ_e) tortuosity with free gas loading for CH_4 and CO_2 in rigid and flexible matrices, respectively. Notably, the consistent τ_g values (indicated by the pink dashed lines) across various gas loadings in a rigid matrix validate its effectiveness in representing the complexity and winding nature of diffusive pathways. For the same rigid matrix, CO_2 shows a slightly higher τ_g (~ 2.49) than CH_4 (~ 2.15), as each molecule “perceives” the pore network differently. This difference arises because molecule size, shape, and collision parameters determine which parts of the pore space are accessible. CO_2 , due to its linear geometry, may encounter more constrictions and adopt more winding routes, resulting in a higher τ_g . In contrast, CH_4 , modeled as a united atom in the TraPPE-UA framework, exhibits an effective compactness and symmetry that facilitates a relatively less tortuous diffusion pathway through the pore structure compared to the linear elongated structure of CO_2 . Moreover, extensive comparisons between united-atom and all-atom models for CH_4 have consistently demonstrated negligible differences in predicting its thermodynamic and dynamics properties under the conditions investigated in this study.^{63,64}

Another noteworthy observation is the consistent decrease in τ_e across all cases. At low loadings, gas molecules tend to interact extensively with pore walls, forcing them to follow more circuitous pathways. As loading increases, adsorption sites become increasingly occupied, causing additional molecules to spend more time in the free or near-free phase. This effectively “shortens” the average path length, leading to a decrease in τ_e . Although this suggests a less tortuous path at

higher loadings, the increased molecular crowding within the same pore space raises the frequency of gas–gas collisions, which in turn impedes overall molecular motions. In other words, while the local “route” becomes less winding, the “traffic” within the pores becomes heavier. This congestion lowers the net D_s , as seen in Figure 4, since each molecule’s motion is increasingly hindered by neighboring molecules. Therefore, the simultaneous decreases in τ_e and D_s are not contradictory but rather complementary reflections of these dynamics.

It is observed that τ_g is larger than τ_e for CO_2 , but the opposite is true for CH_4 . This is due to the anisotropic charge distribution of CO_2 , where orientation plays a significant role during collisions.^{65–67} In the presence of attractive interactions (τ_e), CO_2 can transiently adsorb along surfaces, creating “guideposts” that smooth out its overall path. This orientation-specific “shortcut” effect (surface diffusion and partial ordering) reduces the real path length for CO_2 relative to the geometry-only scenario (τ_g). Conversely, CH_4 , being symmetric and nonpolar, experiences a more isotropic repulsive interaction when attractions are turned off. In such cases, CH_4 diffuses more freely, similar to bulk gas in a confined space, resulting in $\tau_e > \tau_g$.

Considering flexibility, a divergence and convergence trend between τ_g and τ_e emerges for CO_2 and CH_4 , respectively. For CO_2 , strong gas–solid interactions at higher loadings force molecules to follow more efficient pathways along the pore surfaces. This “guiding effect” intensifies with loading, causing τ_e to decrease relative to τ_g , leading to divergence. In contrast, CH_4 ’s weaker interaction with the coal matrix provides fewer advantages at higher loadings. Its motion gradually transitions

toward a bulk-like regime, resulting in the convergence between τ_e and τ_g .

In the rigid matrix, CH_4 maintains a consistent τ_g (~ 2.15), while CO_2 exhibits a slightly higher value (~ 2.49) due to its linear geometry and orientation effects within the same pore structure. When flexibility is introduced (see Figure 7(b) and (d)), both gases initially show increased τ_g (2.80 and 3.20 for CH_4 and CO_2 , respectively) as structural rearrangements in the matrix create more convoluted pathways. However, as loading increases, CH_4 gradually “smoothes” these pathways, lowering its τ_g to ~ 2.40 , which remains slightly above the rigid value (~ 2.15). In contrast, CO_2 induces additional structural distortions at higher loadings, increasing its tortuosity from 3.20 up to ~ 3.70 , significantly exceeding the rigid value of 2.49. This divergence highlights that flexibility can either mitigate or amplify pore complexity, depending on the strength of interactions between gas and host framework as well as the extent to which gas reshapes the matrix.

4. CONCLUSIONS

By performing hybrid GCMC/NPT and EMD simulations, novel insights into the coupled effects of gas adsorption and matrix deformation on gas diffusion in deformable coal matrices are provided, with a particular focus on the differences between CH_4 and CO_2 . The enhanced adsorption observed in flexible coal matrices underscores the paramount role of matrix deformation. CO_2 , driven by its higher quadrupole moment and stronger gas–matrix interactions, exhibits higher adsorption amount and induces greater volumetric swelling than CH_4 , significantly increasing accessible pore volume and limiting pore diameter. Matrix flexibility alleviates the reduction in self-diffusion coefficients caused by increasing pressure and gas loading; specifically, the dependence of CH_4 diffusion on free volume decreases by over 50%, while that of CO_2 decreases by approximately 30%. A universal linear relationship applies to both gases and matrix types, simplifying diffusion modeling while providing reasonable accuracy and practical relevance.

A novel aspect of this work is its examination of the tortuosity effects on gas diffusion. A contrasting comparison between geometrical and effective tortuosity reveals that not all interactions that appear to slow molecular motion are purely detrimental to the mobility. In some cases, slight attractions or orientation-dependent interactions impose a kind of order on the system, increasing the transport efficiency. For CO_2 , attractive interactions with the solid are not just “sticky traps” but contribute to a subtle interplay among orientation, energy landscapes, and pore geometry. These interactions collectively create a more efficient, though seemingly slower, transport pathway. Without them, CO_2 diffusion becomes more random and collision-dominated and ultimately less efficient overall.

The significance of this work lies in its ability to quantitatively link matrix deformation to changes in gas diffusion behavior, providing valuable insights for optimizing the CBM recovery. The findings indicate that matrix flexibility can alleviate reductions in gas self-diffusion coefficients, highlighting a potential approach to enhancing gas mobility under high-pressure conditions crucial for sustained CH_4 production. Furthermore, the observed universal linear correlation between diffusion and the free volume ratio simplifies predictive modeling, establishing a robust framework for estimating gas transport in flexible coal matrices. While this study emphasizes these fundamental relationships, a comprehensive evaluation of how flexibility and gas loading influence

adsorption selectivity (CH_4/CO_2) remains an important future research direction. Additionally, the orientation-dependent diffusion pathways identified for CO_2 (attributed to electrostatic interactions with pore walls as discussed earlier) warrant further systematic investigation through controlled simulations. Subsequent studies will aim to integrate these aspects, bridging the gap between atomistic simulations and engineering applications and ultimately advancing the understanding of gas transport phenomena in complex porous media.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.5c00430>.

Illustration of coal matrix model building and relaxation process, validation of gas force field, illustration of intermittent Brownian motion of a single molecule, validation of free volume theory (DOCX)

■ AUTHOR INFORMATION

Corresponding Authors

Quanlin Yang – College of Safety Science and Engineering and Key Laboratory of Western Mine Exploitation and Hazard Prevention, Ministry of Education, Xi'an University of Science and Technology, Xi'an, Shaanxi 710054, China; School of Mining and Petroleum Engineering, Department of Civil and Environmental Engineering, University of Alberta, Edmonton, AB T6G 1H9, Canada; Email: quanlinyang@xust.edu.cn

Zhehui Jin – School of Mining and Petroleum Engineering, Department of Civil and Environmental Engineering, University of Alberta, Edmonton, AB T6G 1H9, Canada; orcid.org/0000-0001-8305-5637; Email: zhehui2@ualberta.ca

Authors

Junhua Xue – College of Safety Science and Engineering and Key Laboratory of Western Mine Exploitation and Hazard Prevention, Ministry of Education, Xi'an University of Science and Technology, Xi'an, Shaanxi 710054, China

Haifei Lin – College of Safety Science and Engineering and Key Laboratory of Western Mine Exploitation and Hazard Prevention, Ministry of Education, Xi'an University of Science and Technology, Xi'an, Shaanxi 710054, China

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.langmuir.5c00430>

Notes

The authors declare no competing financial interest.

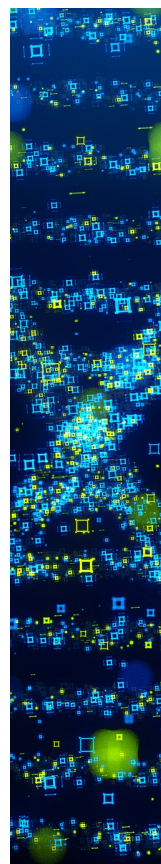
■ ACKNOWLEDGMENTS

The authors acknowledge the China Scholarship Council (CSC) for the financial support provided to Quanlin Yang (No. 202208610140). This research was enabled in part by support provided by the Digital Research Alliance of Canada (<https://alliancecan.ca/en>). The authors also greatly acknowledge a Discovery Grant from the Natural Sciences and Engineering Research Council of Canada (NSERC RGPIN-2023-03587) and Xi'an University of Science and Technology Outstanding Doctoral Dissertation Cultivation Program (Grant No. PY22004).

REFERENCES

- (1) Zhao, W.; Cheng, Y.; Pan, Z.; Wang, K.; Liu, S. Gas diffusion in coal particles: A review of mathematical models and their applications. *Fuel* **2019**, *252*, 77–100.
- (2) Zhang, X. G.; Ranjith, P. G.; Perera, M. S. A.; Ranathunga, A. S.; Haque, A. Gas Transportation and Enhanced Coalbed Methane Recovery Processes in Deep Coal Seams: A Review. *Energy Fuels* **2016**, *30* (11), 8832–8849.
- (3) Falk, K.; Coasne, B.; Pellenq, R.; Ulm, F.-J.; Bocquet, L. Subcontinuum mass transport of condensed hydrocarbons in nanoporous media. *Nat. Commun.* **2015**, *6* (1), 6949.
- (4) Bousige, C.; Levitz, P.; Coasne, B. Bridging scales in disordered porous media by mapping molecular dynamics onto intermittent Brownian motion. *Nat. Commun.* **2021**, *12* (1), 1043.
- (5) Bai, Y.; Lin, H.-F.; Li, S.-G.; Long, H.; Yan, M.; Li, Y.; Qin, L.; Zhou, B. Experimental study on kinetic characteristics of gas diffusion in coal under nitrogen injection. *Energy* **2022**, *254*, No. 124251.
- (6) Meng, Y.; Li, Z. Experimental study on diffusion property of methane gas in coal and its influencing factors. *Fuel* **2016**, *185*, 219–228.
- (7) Liu, S.; Harpalani, S. A new theoretical approach to model sorption-induced coal shrinkage or swelling. *AAPG bulletin* **2013**, *97* (7), 1033–1049.
- (8) Sampath, K. H. S. M.; Perera, M. S. A.; Matthai, S. K.; Ranjith, P. G.; Dong-yin, L. Modelling of fully-coupled CO₂ diffusion and adsorption-induced coal matrix swelling. *Fuel* **2020**, *262*, No. 116486.
- (9) Liu, J.; Fokker, P. A.; Spiers, C. J. Coupling of swelling, internal stress evolution, and diffusion in coal matrix material during exposure to methane. *Journal of Geophysical Research: Solid Earth* **2017**, *122* (2), 844–865.
- (10) Zhang, H.; Diao, R.; Chan, H. H.; Mostofi, M.; Evans, B. Molecular simulation of the adsorption-induced deformation during CO₂ sequestration in shale and coal carbon slit pores. *Fuel* **2020**, *272*, No. 117693.
- (11) Zhang, J.; Liu, K.; Clennell, M. B.; Dewhurst, D. N.; Pervukhina, M. Molecular simulation of CO₂–CH₄ competitive adsorption and induced coal swelling. *Fuel* **2015**, *160*, 309–317.
- (12) Yang, Q.; Xue, J.; Lin, H.; Jin, Z. Two-way coupling dynamics of CH₄ adsorption and coal matrix deformation: Insights from hybrid GCMC/MD simulations. *Chemical Engineering Journal* **2024**, *498*, No. 155321.
- (13) Zhang, Q.; Zhu, H. Macromolecular insights into the influence of bituminous coal matrix deformation on CH₄–N₂ competitive adsorption and diffusion. *Colloids Surf., A* **2024**, *694*, No. 134143.
- (14) Yang, Q.; Xue, J.; Li, W.; Hu, B.; Ma, Q.; Zhan, K.; Du, X.; Chen, Z. Reconstructions of supercritical CO₂ adsorption isotherms and absolute adsorption estimation in nanoporous coals considering volumetric effects and varying adsorbed phase densities. *Chemical Engineering Journal* **2022**, *433*, No. 133492.
- (15) Hu, B.; Cheng, Y.; He, X.; Wang, Z.; Jiang, Z.; Wang, C.; Li, W.; Wang, L. New insights into the CH₄ adsorption capacity of coal based on microscopic pore properties. *Fuel* **2020**, *262*, No. 116675.
- (16) Obliger, A.; Ulm, F.-J.; Pellenq, R. Impact of nanoporosity on hydrocarbon transport in shales' organic matter. *Nano Lett.* **2018**, *18* (2), 832–837.
- (17) Obliger, A.; Pellenq, R.; Ulm, F.-J.; Coasne, B. Free volume theory of hydrocarbon mixture transport in nanoporous materials. *J. Phys. Chem. Lett.* **2016**, *7* (19), 3712–3717.
- (18) Ariskina, K.; Galliéro, G.; Obliger, A. Free volume model for transport in flexible kerogen of source rock's organic matter. *J. Phys. Chem. B* **2022**, *126* (38), 7409–7417.
- (19) Obliger, A.; Bousige, C.; Coasne, B.; Leyssale, J.-M. Development of Atomistic Kerogen Models and Their Applications for Gas Adsorption and Diffusion: A Mini-Review. *Energy Fuels* **2023**, *37* (3), 1678–1698.
- (20) Schlaich, A.; Barrat, J.-L.; Coasne, B. Theory and Modeling of Transport for Simple Fluids in Nanoporous Materials: From Microscopic to Coarse-Grained Descriptions. *Chem. Rev.* **2025**, *125* (5), 2561–2624.
- (21) Ariskina, K.; Galliéro, G.; Obliger, A. Adsorption-induced swelling impact on CO₂ transport in kerogen microporosity described by free volume theory. *Fuel* **2024**, *359*, No. 130475.
- (22) Moulτος, O. A.; Zhang, Y.; Tsimpanogiannis, I. N.; Economou, I. G.; Maginn, E. J. System-size corrections for self-diffusion coefficients calculated from molecular dynamics simulations: The case of CO₂, n-alkanes, and poly(ethylene glycol) dimethyl ethers. *J. Chem. Phys.* **2016**, *145* (7), No. 074109.
- (23) Collell, J.; Galliero, G.; Vermorel, R.; Ungerer, P.; Yiannourakou, M.; Montel, F.; Pujol, M. Transport of multi-component hydrocarbon mixtures in shale organic matter by molecular simulations. *J. Phys. Chem. C* **2015**, *119* (39), 22587–22595.
- (24) Thyagarajan, R.; Sholl, D. S. Molecular simulations of CH₄ and CO₂ diffusion in rigid nanoporous amorphous materials. *J. Phys. Chem. C* **2022**, *126* (19), 8530–8538.
- (25) Yu, K. B.; Bowers, G. M.; Loganathan, N.; Kalinichev, A. G.; Yazaydin, A. O. Diffusion behavior of methane in 3D kerogen models. *Energy Fuels* **2021**, *35* (20), 16515–16526.
- (26) Aljaberi, J.; Alafnan, S.; Glatz, G.; Sultan, A. S.; Afagwu, C. The impact of kerogen tortuosity on shale permeability. *SPE Journal* **2021**, *26* (02), 765–779.
- (27) Vasileiadis, M.; Peristeras, L. D.; Papavasileiou, K. D.; Economou, I. G. Transport properties of shale gas in relation to kerogen porosity. *J. Phys. Chem. C* **2018**, *122* (11), 6166–6177.
- (28) Li, W.; Nan, Y.; Jin, Z. Dependence of methane transport on pore informatics in the amorphous nanoporous kerogen matrix. *Langmuir* **2024**, *40* (1), 687–695.
- (29) Wu, T.; Firoozabadi, A. Effect of microstructural flexibility on methane flow in kerogen matrix by molecular dynamics simulations. *J. Phys. Chem. C* **2019**, *123* (17), 10874–10880.
- (30) Obliger, A.; Valdenaire, P.-L.; Capit, N.; Ulm, F. J.; Pellenq, R. J. M.; Leyssale, J.-M. Poroelasticity of methane-loaded mature and immature kerogen from molecular simulations. *Langmuir* **2018**, *34* (45), 13766–13780.
- (31) Obliger, A.; Valdenaire, P.-L.; Ulm, F.-J.; Pellenq, R. J. M.; Leyssale, J.-M. Methane diffusion in a flexible kerogen matrix. *J. Phys. Chem. B* **2019**, *123* (26), 5635–5640.
- (32) Potier, K.; Ariskina, K.; Obliger, A.; Leyssale, J.-M. Molecular simulation of Argon adsorption and diffusion in a microporous carbon with poroelastic couplings. *Langmuir* **2023**, *39* (27), 9384–9395.
- (33) Yang, Y.; Narayanan Nair, A. K.; Sun, S. Sorption and diffusion of methane and carbon dioxide in amorphous poly(alkyl acrylates): A molecular simulation study. *J. Phys. Chem. B* **2020**, *124* (7), 1301–1310.
- (34) Ungerer, P.; Collell, J.; Yiannourakou, M. Molecular modeling of the volumetric and thermodynamic properties of kerogen: Influence of organic type and maturity. *Energy Fuels* **2015**, *29* (1), 91–105.
- (35) Kelemen, S. R.; Afeworki, M.; Gorbaty, M. L.; Sansone, M.; Kwiatek, P. J.; Walters, C. C.; Freund, H.; Siskin, M.; Bence, A. E.; Curry, D. J.; et al. Direct characterization of kerogen by X-ray and solid-state ¹³C nuclear magnetic resonance methods. *Energy Fuels* **2007**, *21* (3), 1548–1561.
- (36) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. PACKMOL: A package for building initial configurations for molecular dynamics simulations. *J. Comput. Chem.* **2009**, *30* (13), 2157–2164.
- (37) He, J.; Ju, Y.; Lammers, L.; Kulasinski, K.; Zheng, L. Tortuosity of kerogen pore structure to gas diffusion at molecular- and nano-scales: A molecular dynamics simulation. *Chem. Eng. Sci.* **2020**, *215*, No. 115460.
- (38) Michalec, L.; Lísál, M. Molecular simulation of shale gas adsorption onto overmature type II model kerogen with control microporosity. *Mol. Phys.* **2017**, *115* (9–12), 1086–1103.
- (39) Stukowski, A. Visualization and analysis of atomistic simulation data with OVITO—the open visualization tool. *Modell. Simul. Mater. Sci. Eng.* **2010**, *18* (1), No. 015012.

- (40) Dauber-Osguthorpe, P.; Roberts, V. A.; Osguthorpe, D. J.; Wolff, J.; Genest, M.; Hagler, A. T. Structure and energetics of ligand binding to proteins: Escherichia coli dihydrofolate reductase-trimethoprim, a drug-receptor system. *Proteins: Struct., Funct., Bioinf.* **1988**, *4* (1), 31–47.
- (41) Maple, J. R.; Dinur, U.; Hagler, A. T. Derivation of force fields for molecular mechanics and dynamics from ab initio energy surfaces. *Proc. Natl. Acad. Sci. U. S. A.* **1988**, *85* (15), 5350–5354.
- (42) Martin, M. G.; Siepmann, J. I. Transferable potentials for phase equilibria. I. United-atom description of n-Alkanes. *J. Phys. Chem. B* **1998**, *102* (14), 2569–2577.
- (43) Potoff, J. J.; Siepmann, J. I. Vapor–liquid equilibria of mixtures containing alkanes, carbon dioxide, and nitrogen. *AIChE J.* **2001**, *47* (7), 1676–1682.
- (44) Kamberaj, H.; Low, R. J.; Neal, M. P. Time reversible and symplectic integrators for molecular dynamics simulations of rigid molecules. *J. Chem. Phys.* **2005**, *122* (22), No. 224114.
- (45) Sun, H. COMPASS: An ab Initio Force-Field Optimized for Condensed-Phase Applications Overview with Details on Alkane and Benzene Compounds. *J. Phys. Chem. B* **1998**, *102* (38), 7338–7364.
- (46) Thompson, A. P.; Aktulga, H. M.; Berger, R.; Bolintineanu, D. S.; Brown, W. M.; Crozier, P. S.; in 't Veld, P. J.; Kohlmeyer, A.; Moore, S. G.; Nguyen, T. D.; et al. LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **2022**, *271*, No. 108171.
- (47) Hol, S.; Gensterblum, Y.; Massarotto, P. Sorption and changes in bulk modulus of coal — experimental evidence and governing mechanisms for CBM and ECBM applications. *International Journal of Coal Geology* **2014**, *128*–129, 119–133.
- (48) Weishauptová, Z.; Přibyl, O.; Šýkorová, I.; Machovič, V. Effect of bituminous coal properties on carbon dioxide and methane high pressure sorption. *Fuel* **2015**, *139*, 115–124.
- (49) Corrente, N. J.; Parashar, S.; Gough, R.; Hinks, E. L.; Ravikovitch, P. I.; Neimark, A. V. Modeling structural flexibility in 3D carbon models: A hybrid MC/MD approach to adsorption-induced deformation. *Carbon* **2025**, *238*, No. 120160.
- (50) Frenkel, D.; Smit, B. Chapter 8 - Free-energy calculations. In *Understanding Molecular Simulation (Third ed.)*, Frenkel, D., Smit, B., Eds.; Academic Press, 2023; pp 263–321.
- (51) Martin, M. G. MCCCSTowhee: a tool for Monte Carlo molecular simulation. *Mol. Simul.* **2013**, *39* (14–15), 1212–1222.
- (52) Jamali, S. H.; Wolff, L.; Becker, T. M.; de Groen, M.; Ramdin, M.; Hartkamp, R.; Bardow, A.; Vlugt, T. J. H.; Moulton, O. A. OCTP: A tool for on-the-fly calculation of transport properties of fluids with the order-n algorithm in LAMMPS. *J. Chem. Inf. Model.* **2019**, *59* (4), 1290–1294.
- (53) Maginn, E. J.; Messerly, R. A.; Carlson, D. J.; Roe, D. R.; Elliot, J. R. Best practices for computing transport properties 1. Self-diffusivity and viscosity from equilibrium molecular dynamics. *Living Journal of Computational Molecular Science* **2018**, *1* (1), 6324.
- (54) Frenkel, D.; Smit, B. Chapter 5 - Computer experiments. In *Understanding Molecular Simulation (Third ed.)*; Academic Press, 2023; pp 125–177.
- (55) Hammonds, K. D.; Heyes, D. M. Shadow Hamiltonian in classical NVE molecular dynamics simulations involving Coulomb interactions. *J. Chem. Phys.* **2021**, *154* (17), No. 174102.
- (56) Smit, B.; et al. Molecular Simulations of Zeolites: Adsorption, Diffusion, and Shape Selectivity. *Chem. Rev.* **2008**, *108* (10), 4125–4184.
- (57) Sarkisov, L.; Bueno-Perez, R.; Sutharson, M.; Fairen-Jimenez, D. Materials informatics with PoreBlazer v4.0 and the CSD MOF database. *Chem. Mater.* **2020**, *32* (23), 9849–9867.
- (58) Elwinger, F.; Pourmand, P.; Furó, I. Diffusive transport in pores. Tortuosity and molecular interaction with the pore wall. *J. Phys. Chem. C* **2017**, *121* (25), 13757–13764.
- (59) Ghanbarian, B.; Hunt, A. G.; Ewing, R. P.; Sahimi, M. Tortuosity in Porous Media: A Critical Review. *Soil Science Society of America Journal* **2013**, *77* (5), 1461–1477.
- (60) Zhang, C.; Shomali, A.; Coasne, B.; Derome, D.; Carmeliet, J. Sorption–deformation–percolation model for diffusion in nanoporous media. *ACS Nano* **2023**, *17* (5), 4507–4514.
- (61) Kulasinski, K.; Guyer, R.; Derome, D.; Carmeliet, J. Water diffusion in amorphous hydrophilic systems: A stop and go process. *Langmuir* **2015**, *31* (39), 10843–10849.
- (62) Kellouai, W.; Judeinstein, P.; Plazenet, M.; Zanotti, J.-M.; Berrod, Q.; Drobek, M.; Julbe, A.; Coasne, B. Free volume theory of self-diffusion in zeolites: Molecular simulation and experiment. *Microporous Mesoporous Mater.* **2025**, *381*, No. 113305.
- (63) Aimoli, C. G.; Maginn, E. J.; Abreu, C. R. A. Transport properties of carbon dioxide and methane from molecular dynamics simulations. *J. Chem. Phys.* **2014**, *141* (13), No. 134101.
- (64) dos Santos, T. J. P.; Abreu, C. R. A.; Horta, B. A. C.; Tavares, F. W. Self-diffusion coefficients of methane/n-hexane mixtures at high pressures: An evaluation of the finite-size effect and a comparison of force fields. *Journal of Supercritical Fluids* **2020**, *155*, No. 104639.
- (65) Bowers, G. M.; Schaefer, H. T.; Loring, J. S.; Hoyt, D. W.; Burton, S. D.; Walter, E. D.; Kirkpatrick, R. J. Role of Cations in CO₂ Adsorption, Dynamics, and Hydration in Smectite Clays under In Situ Supercritical CO₂ Conditions. *J. Phys. Chem. C* **2017**, *121* (1), 577–592.
- (66) Svoboda, M.; Brennan, J. K.; Lissal, M. Molecular dynamics simulation of carbon dioxide in single-walled carbon nanotubes in the presence of water: structure and diffusion studies. *Mol. Phys.* **2015**, *113* (9–10), 1124–1136.
- (67) Takaba, H.; Mizukami, K.; Kubo, M.; Fahmi, A.; Miyamoto, A. Permeation dynamics of small molecules through silica membranes: Molecular dynamics study. *AIChE J.* **1998**, *44* (6), 1335–1343.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the
right biological insights in seconds

Start your search



A division of the
American Chemical Society